

of implementing a discrete-time state estimator. Suppose that we have a system given as

$$\begin{aligned}\dot{x} &= Ax + Bu + w \\ y &= Cx + v \\ w &\sim (0, Q_c) \\ v &\sim (0, R_c)\end{aligned}\tag{8.19}$$

Both $w(t)$ and $v(t)$ are continuous-time noise, and $u(t)$ is a known input. This system is approximately equivalent to the following discrete-time system:

$$\begin{aligned}x_k &= e^{A\Delta t}x_{k-1} + e^{A\Delta t} \int_0^{\Delta t} e^{-A\tau} d\tau Bu_{k-1} + w_k \\ &= e^{A\Delta t}x_{k-1} + e^{A\Delta t} [I - e^{-A\Delta t}] A^{-1} Bu_{k-1} + w_k \\ y_k &= Cx_k + v_k \\ w_k &\sim (0, Q_c\Delta t) \\ v_k &\sim (0, R_c/\Delta t)\end{aligned}\tag{8.20}$$

where Δt is the discretization step size. The second expression for x_k above is valid if A^{-1} exists. If we use these discretized equations to simulate a continuous-time system, then we can simulate a continuous-time state estimator using the resulting measurements with one of the integration methods discussed in Section 1.5. The remainder of this chapter discusses continuous-time state estimation.

8.2 DERIVATION OF THE CONTINUOUS-TIME KALMAN FILTER

We will now use the results of the previous section to derive the continuous-time Kalman filter. Suppose that we have a continuous-time system given as

$$\begin{aligned}\dot{x} &= Ax + Bu + w \\ y &= Cx + v \\ w &\sim (0, Q_c) \\ v &\sim (0, R_c)\end{aligned}\tag{8.21}$$

When we write $w \sim (0, Q_c)$ we mean exactly what is written in Equation (8.11). When we write $v \sim (0, R_c)$ we mean exactly what is written in Equation (8.18). Now suppose that we discretize this system with a sample time of T (see Section 1.4). We obtain

$$\begin{aligned}x_k &= Fx_{k-1} + Gu_{k-1} + \Lambda w_{k-1} \\ y_k &= Hx_k + v_k\end{aligned}\tag{8.22}$$

The matrices in this discrete-time system are computed as follows:

$$\begin{aligned}
F &= \exp(AT) \\
&\approx (I + AT) \text{ for small } T \\
G &= (\exp(AT) - I)A^{-1}B \\
&\approx BT \text{ for small } T \\
\Lambda &= (\exp(AT) - I)A^{-1} \\
&\approx IT \text{ for small } T \\
H &= C \\
w_k &\sim (0, Q), \quad Q = Q_c T \\
v_k &\sim N(0, R), \quad R = R_c/T
\end{aligned} \tag{8.23}$$

The discrete-time Kalman filter gain for this system was derived in Section 5.1 as

$$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1} \tag{8.24}$$

From this we can derive

$$\begin{aligned}
K_k &= P_k^- C^T (C P_k^- C^T + R_c/T)^{-1} \\
\frac{K_k}{T} &= P_k^- C^T (C P_k^- C^T T + R_c)^{-1} \\
\lim_{T \rightarrow 0} \frac{K_k}{T} &= P_k^- C^T R_c^{-1}
\end{aligned} \tag{8.25}$$

The estimation-error covariances were derived in Section 5.1 as

$$\begin{aligned}
P_k^+ &= (I - K_k H) P_k^- \\
P_{k+1}^- &= F P_k^+ F^T + Q
\end{aligned} \tag{8.26}$$

For small values of T , this can be written as

$$\begin{aligned}
P_{k+1}^- &= (I + AT) P_k^+ (I + AT)^T + Q_c T \\
&= P_k^+ + (A P_k^+ + P_k^+ A^T + Q_c) T + A P_k^+ A^T T^2
\end{aligned} \tag{8.27}$$

Substituting for P_k^+ gives

$$\begin{aligned}
P_{k+1}^- &= (I - K_k C) P_k^- + A P_k^+ A^T T^2 + \\
&\quad [A(I - K_k C) P_k^- + (I - K_k C) P_k^- A^T + Q_c] T
\end{aligned} \tag{8.28}$$

Subtracting P_k^- from both sides and then dividing by T gives

$$\begin{aligned}
\frac{P_{k+1}^- - P_k^-}{T} &= \frac{-K_k C P_k^-}{T} + A P_k^+ A^T T + \\
&\quad (A P_k^- + A K_k C P_k^- + P_k^- A^T - K_k C P_k^- A^T + Q_c)
\end{aligned} \tag{8.29}$$

Taking the limit as $T \rightarrow 0$ and using Equation (8.25) gives

$$\begin{aligned}
\dot{P} &= \lim_{T \rightarrow 0} \frac{P_{k+1}^- - P_k^-}{T} \\
&= -P C^T R_c^{-1} C P + A P + P A^T + Q_c
\end{aligned} \tag{8.30}$$

This equation for P is called a differential Riccati equation and can be used to compute the estimation-error covariance for the continuous-time Kalman filter. This requires n^2 integrations because P is an $n \times n$ matrix. But P is symmetric, so in practice we only need to integrate $n(n+1)/2$ equations in order to solve for P .

In Section 5.1 we derived the Kalman filter equations for $\hat{x}$ as

$$\begin{aligned}\hat{x}_k^- &= F\hat{x}_{k-1}^+ + Gu_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H\hat{x}_k^-)\end{aligned}\quad (8.31)$$

If we assume that T is small we can use Equation (8.23) to write the measurement update equation as

$$\begin{aligned}\hat{x}_k^+ &= F\hat{x}_{k-1}^+ + Gu_{k-1} + K_k(y_k - HF\hat{x}_{k-1}^+ - HGu_{k-1}) \\ &\approx (I + AT)\hat{x}_{k-1}^+ + BTu_{k-1} + \\ &\quad K_k(y_k - C(I + AT)\hat{x}_{k-1}^+ - CBTu_{k-1})\end{aligned}\quad (8.32)$$

Now substitute for K_k from Equation (8.25) to obtain

$$\begin{aligned}\hat{x}_k^+ &= \hat{x}_{k-1}^+ + AT\hat{x}_{k-1}^+ + BTu_{k-1} + \\ &\quad PC^TR_c^{-1}T(y_k - C\hat{x}_{k-1}^+ - CAT\hat{x}_{k-1}^+ - CBTu_{k-1})\end{aligned}\quad (8.33)$$

Subtracting $\hat{x}_{k-1}^+$ from both sides, dividing by T , and taking the limit as $T \rightarrow 0$, gives

$$\begin{aligned}\dot{\hat{x}} &= \lim_{T \rightarrow 0} \frac{\hat{x}_k^+ - \hat{x}_{k-1}^+}{T} \\ &= A\hat{x} + Bu + PC^TR_c^{-1}(y - C\hat{x})\end{aligned}\quad (8.34)$$

This can be written as

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\ K &= PC^TR_c^{-1}\end{aligned}\quad (8.35)$$

This gives the differential equation that can be used to integrate the state estimate in the continuous-time Kalman filter.

The continuous-time Kalman filter

The continuous-time Kalman filter can be summarized as follows.

1. The continuous-time system dynamics and measurement equations are given as

$$\begin{aligned}\dot{x} &= Ax + Bu + w \\ y &= Cx + v \\ w &\sim (0, Q_c) \\ v &\sim (0, R_c)\end{aligned}\quad (8.36)$$

Note that $w(t)$ and $v(t)$ are continuous-time white noise processes.

2. The continuous-time Kalman filter equations are given as

$$\begin{aligned}
 \hat{x}(0) &= E[x(0)] \\
 P(0) &= E[(x(0) - \hat{x}(0))(x(0) - \hat{x}(0))^T] \\
 K &= PC^T R_c^{-1} \\
 \dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\
 \dot{P} &= -PC^T R_c^{-1} CP + AP + PA^T + Q_c
 \end{aligned} \tag{8.37}$$

Other methods of deriving the continuous-time Kalman filter also exist. For example, George Johnson presented a derivation that is based on finding the gain that minimizes the derivative of the estimation covariance [Joh69].

■ EXAMPLE 8.1

In this example we will use the continuous-time Kalman filter to estimate a constant given continuous-time noisy measurements:

$$\begin{aligned}
 \dot{x} &= 0 \\
 y &= x + v \\
 v &\sim (0, R)
 \end{aligned} \tag{8.38}$$

We see that $A = 0$, $Q = 0$, and $C = 1$. Equation (8.37) gives the differential equation for the covariance as

$$\begin{aligned}
 \dot{P} &= -PC^T R^{-1} CP + AP + PA^T + Q \\
 &= -P^2/R
 \end{aligned} \tag{8.39}$$

with the initial condition $P(0) = P_0$. From this we can derive

$$\begin{aligned}
 \frac{dP}{P^2} &= \frac{-d\tau}{R} \\
 \int_{P(0)}^{P(t)} \frac{1}{P^2} dP &= -\int_0^t \frac{1}{R} d\tau \\
 -(P^{-1} - P_0^{-1}) &= -t/R \\
 P^{-1} &= P_0^{-1} + t/R \\
 P &= (P_0^{-1} + t/R)^{-1} \\
 &= \frac{P_0}{1 + P_0 t/R} \\
 \lim_{t \rightarrow \infty} P &= 0
 \end{aligned} \tag{8.40}$$

Equation (8.37) gives the Kalman gain as

$$\begin{aligned}
 K &= PC^T R^{-1} \\
 &= \frac{P_0/R}{1 + P_0 t/R} \\
 \lim_{t \rightarrow \infty} K &= 0
 \end{aligned} \tag{8.41}$$

Equation (8.37) gives the state-update equation as

$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x}) \quad (8.42)$$

from which we can derive

$$\begin{aligned} \dot{\hat{x}} &= K(y - \hat{x}) \\ \lim_{t \rightarrow \infty} \dot{\hat{x}} &= 0 \end{aligned} \quad (8.43)$$

This shows that as time goes to infinity, $\hat{x}$ reaches a steady-state value. This is intuitive because as we obtain an infinite number of measurements of a constant, our estimate of that constant becomes perfect and additional measurements cannot improve our estimate. Furthermore, the Kalman gain goes to zero as time goes to infinity, which again says that we ignore additional measurements (since our estimate becomes perfect). Finally, the covariance P goes to zero as time goes to infinity, which says that the uncertainty in our estimate goes to zero, meaning that our estimate is perfect. Compare this example with the equivalent discrete-time system discussed in Example 7.10.

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■ EXAMPLE 8.2

In this example we are able to obtain measurements of the velocity of an object that is moving in one dimension. The object is subject to random accelerations. We want to estimate the velocity x from noisy velocity measurements. The system and measurement equations are given as

$$\begin{aligned} \dot{x} &= w \\ y &= x + v \\ w &\sim (0, Q) \\ v &\sim (0, R) \end{aligned} \quad (8.44)$$

We see that $A = 0$ and $C = 1$. From the covariance update of Equation (8.37) we obtain

$$\begin{aligned} \dot{P} &= -PC^TR^{-1}CP + AP + PA^T + Q \\ &= -P^2/R + Q \end{aligned} \quad (8.45)$$

with the initial condition $P(0) = P_0$. From this we can derive

$$\begin{aligned} dP &= (Q - P^2/R)d\tau \\ \int_{P(0)}^{P(t)} \frac{dP}{Q - P^2/R} &= \int_0^t d\tau \\ \frac{1}{2\sqrt{Q}} \ln \left(\frac{\sqrt{Q} + P/\sqrt{R}}{\sqrt{Q} - P/\sqrt{R}} \right) \Bigg|_{P(0)}^{P(t)} &= t \end{aligned} \quad (8.46)$$

Solving this for P gives

$$\begin{aligned} P &= \sqrt{QR} \left[\frac{P_0 - \sqrt{QR} + (\sqrt{QR} + P_0) \exp(2t\sqrt{Q})}{\sqrt{QR} - P_0 + (\sqrt{QR} + P_0) \exp(2t\sqrt{Q})} \right] \\ \lim_{t \rightarrow \infty} P &= \sqrt{QR} \end{aligned} \quad (8.47)$$